

2-D Lattice Experiment

By: Helder Rebelo

Professor: Xing

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Structure

For this research, I chose a 2-dimensional diamond lattice configuration. This decision was influenced by various factors beyond the simplicity of the structure's design. The diamond lattice provides a distinct blend of simplicity and mechanical effectiveness, rendering it perfect for exploring basic behaviors in metamaterials. Its unique geometry facilitated straightforward adjustments of the model's dimensions via linear scaling, essential for investigating the size effect on lattice behavior while preserving the core design.

The first cell component, acting as the foundation for all following lattices, was created with measurements of 5 units in both height and length, as illustrated in Figure 1. This dimension was selected to guarantee smooth incorporation when constructing larger assemblies

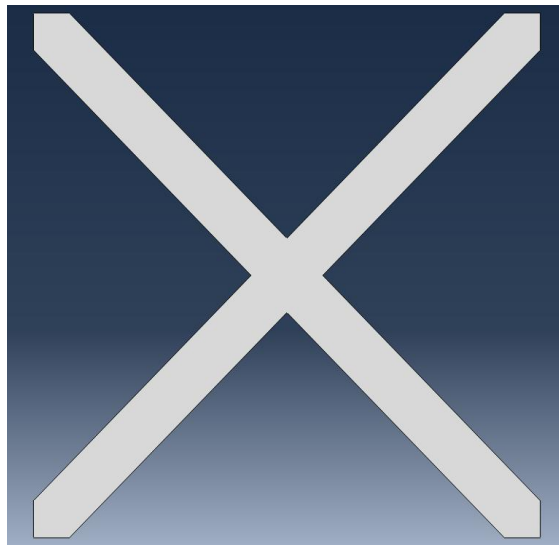


Figure 1: Initial part design

Utilizing this cellular component, I created a 4x4 grid, leading to an assembly design of 20x20 units (Figure 2). This 4x4 grid formed the basis for creating larger grids, such as 8x8 and

16x16 layouts. The capacity to swiftly modify the lattice dimensions by altering the linear configuration greatly simplified the testing procedure.

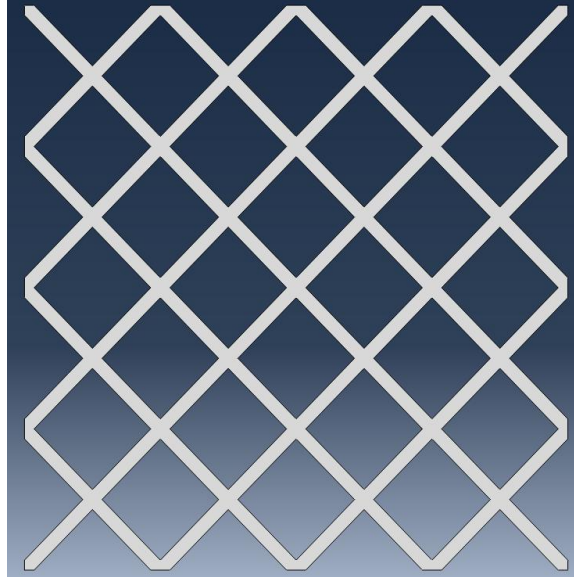
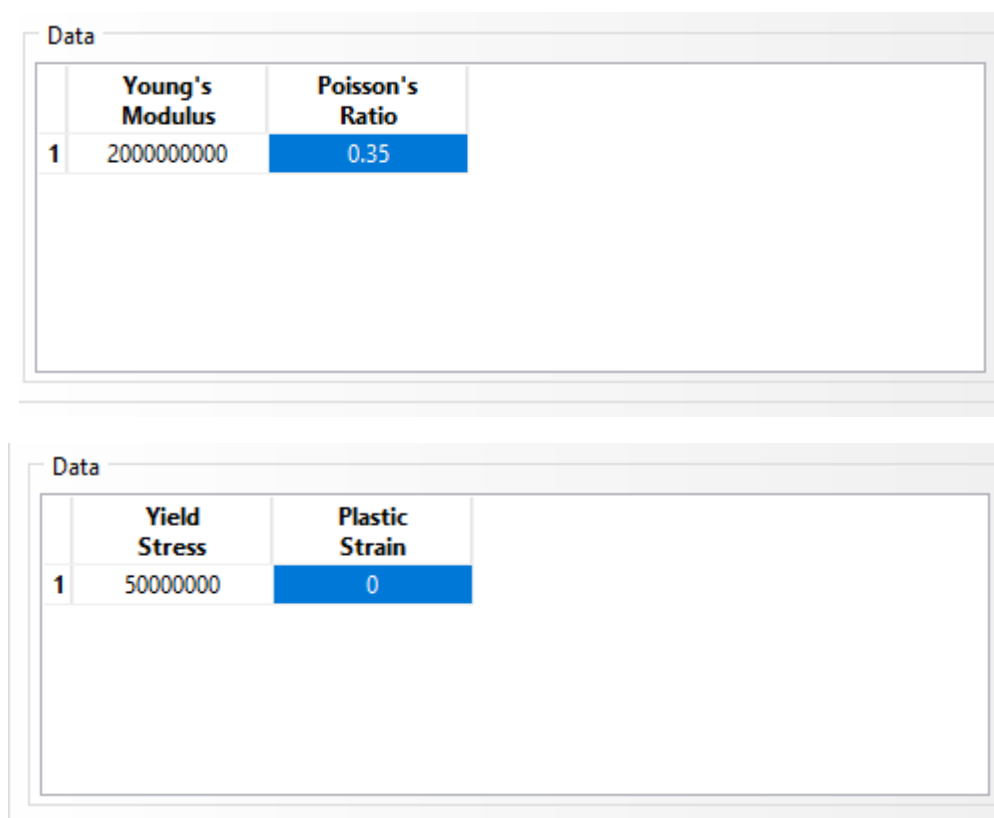


Figure 2: 4x4 cell lattice

The simplicity of altering the lattice size was crucial in performing a comparative analysis on how scale affects the mechanical properties of the lattice. By keeping a uniform design and solely adjusting the dimensions, I could separate and examine the impact of scale on the lattice's performance

Material

The material I chose to study was trimethylolpropane triacrylate. This material is a very commonly used material in lattice structure design because of its various properties and its easy formability. Because of this I believe choosing this material to study would be the most beneficial for our studies. The images below are the material properties that I input into Abaqus to create the trimethylolpropane triacrylate. The young's modulus being $2e9$ pa, poisson's ration 0.35, yield stress $50e6$ Pa and the plastic strain being 0.00. I chose those units and values to make the process of getting stress, strain and displacement information easier.



The figure displays two screenshots of the Abaqus material property input interface. The top screenshot shows the 'Data' table for Young's Modulus and Poisson's Ratio. The bottom screenshot shows the 'Data' table for Yield Stress and Plastic Strain.

	Young's Modulus	Poisson's Ratio
1	2000000000	0.35

	Yield Stress	Plastic Strain
1	50000000	0

Figure 3: Abaqus Material Properties

Parameters

In this experiment, I am employing a diamond 2-dimensional lattice setup with dimensions of 5 units in both height and length. I chose this length because it matched the number of cells I would be using when duplicating the original part to create the 4x4, 8x8, and 16x16 grids. Next, I recreated the initial section, consolidating all the cells into one cohesive unit by merging all the merged cells into a single entity. I built the strong lattice as a homogeneous solid in two stages: the initial conditions and the application phase. I then developed a mesh for the component utilizing an advanced quad element configuration. Fronting utilizing the unbounded approach and a mesh element size of 0.2. I established a reference point at the top center of the grid and restricted it to the entire upper surface to facilitate easier data extraction in the conclusion. I then set a boundary condition throughout the entire base of the lattice to allow no movement in any direction. Upon finishing that, I applied a shift of 2 units in the U2 direction, guided by my reference point, which was confined to the full upper surface of the lattice. Afterward, I produced a history output for RF2 and U2 at the reference point, which gave me the displacement and reaction force readings across 100 points. Using this information, I was able to create the Displacement Force Graphs using Minitab. For the wire beam model, I chose a circular beam section with a radius of 0.1. This decision was influenced by several factors:

- Simplicity: A circular cross-section streamlines the model, lowering computational complexity while effectively capturing the fundamental bending characteristics of beams.
- Isotropic Behavior: Circular sections display isotropic properties, indicating their characteristics are uniform in every direction. This removes the necessity to account for orientation-dependent impacts, further streamlining the analysis.
- Common Application: Circular shapes are frequently utilized in wire-like formations, enhancing the model's representation of practical uses.
- Computational Efficiency: In contrast to more intricate cross-sections, circular beams need fewer computations, resulting in quicker simulation durations

Solid Lattice

In the solid lattice simulations, I found that the stress distribution predominantly exhibits an "X" pattern, with the greatest concentrations present along the primary diagonal areas of the structure. The lattice consists of interlinked diamond-shaped cells, and during loading, the forces are mainly conveyed through these diagonal routes. The regions showing warmer hues—yellow, orange, and red—denote the most strained areas, implying that these parts endure the bulk of the stress.

I also noted that the outer edges and corners undergo considerably reduced stress levels, as indicated by the cooler blue hues. This suggests that the forces exerted are not evenly spread but are instead concentrated through the central diagonal areas. The lattice's reaction to the load indicates that, although deformation takes place, the structure stays stable within the specified displacement limits.

With this increased lattice size, I observed an enhancement in stress dissipation compared to more compact versions of the structure. The extra structural components assist in spreading the load more efficiently, minimizing the intensity of localized stress concentrations. Nonetheless, the junctions of the diagonal components still face the greatest amounts of strain, rendering them crucial points for possible failure under heightened loading circumstances.

4x4 Solid Lattice

In this setup, there is increased stress concentration primarily in the main diagonal regions, which endure the highest forces during deformation. The color gradient shows that the junctions of the diagonal elements experience the greatest forces, particularly in the red regions, suggesting potential critical points for structural collapse. The outer segments, particularly in their centers, experience relatively low stress levels, indicated by the blue and green hues. These areas appear to remain largely unaffected during maximum load conditions, indicating that they do not significantly influence the structural response in this specific loading scenario.

Significantly, despite the applied displacement, the structure bends without displaying any obvious signs of complete failure. However, the areas experiencing the most strain, particularly the bright red zones, reflect regions where the material is facing considerable tension (notably in the upper parts) and compression (in the lower section). These localized stress concentrations suggest that if the load were to rise more, these regions would likely be the first to fail, potentially due to material yielding or fracture. Overall, while deformation takes place, the structure maintains its integrity under the defined conditions; however, careful attention must be directed toward the highly stressed joints to prevent potential failure in prolonged or elevated loading scenarios.

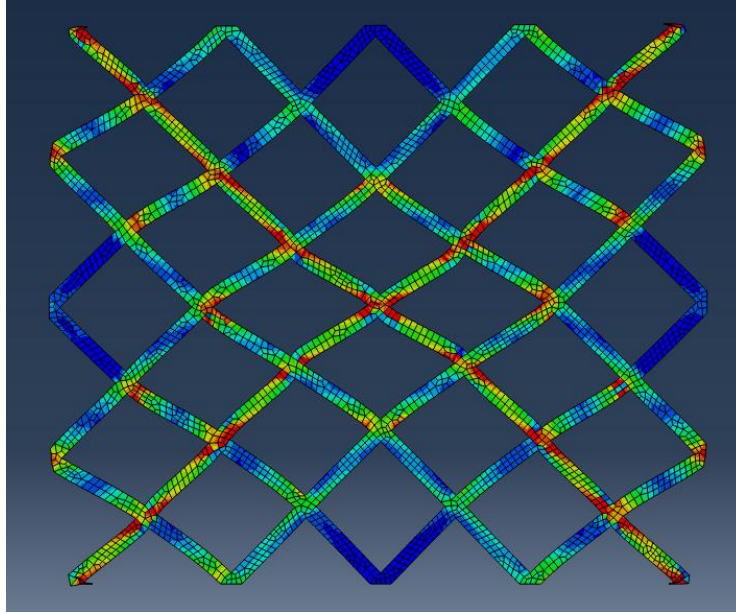


Figure 4: 4x4 Solid Lattice Cell Structure Under Displacement

We can see from the table that from 0 to 0.6 meters the force increases in a linear fashion from 0 to 2.75×10^7 newtons. This area of the graph is the elastic limit, ending at elastic limit at approximately 0.6 meters. After that we enter the plastic zone of the deformation. Because we did not deform the section more, we end the graph on the yield point of the material and never reach the ultimate stress or fracture point of the material.

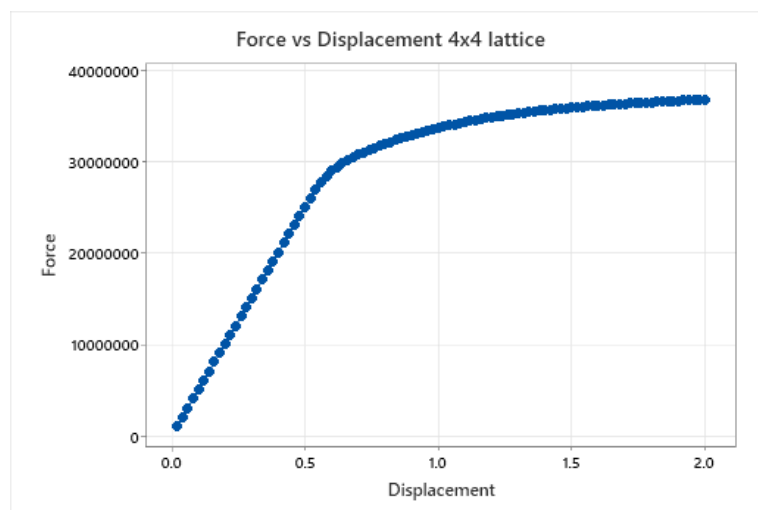


Figure 5: 4x4 Solid Lattice Force vs Displacement Graph

8x8 Solid Lattice

Like the findings observed in the previous 4x4 lattice model, the stress distribution in this larger lattice predominantly shows an X-shaped pattern. This trend is particularly clear in this model as its greater size allows stress to distribute more effectively across the structure. The wider scope provides a fuller view of the force distribution, highlighting the buildup of stress along the main diagonal regions. The diamond-shaped unit cells in the lattice aid in this stress distribution by directing the forces to the central diagonal regions, where the highest stress concentrations take place. These regions experience the greatest amounts of stress and are likely the most vulnerable to structural failure if the applied displacement were to increase. The intersections of these diagonal elements continue to exhibit considerable stress accumulation, similar to the smaller lattice model, identifying these areas as critical points for potential structural problems. Due to the increased lattice size and maintaining the same displacement as in the smaller model, the total deformation effect on the structure is reduced. The additional structural elements help to distribute the applied forces more evenly, leading to decreased localized stress concentrations compared to the smaller lattice. This suggests that more extensive lattice structures of this nature might offer improved load distribution and reduced likelihood of failure in similar displacement situations. However, attention still needs to be given to the high-stress diagonal zones, as they remain the most susceptible to strain-induced deformation or potential failure.

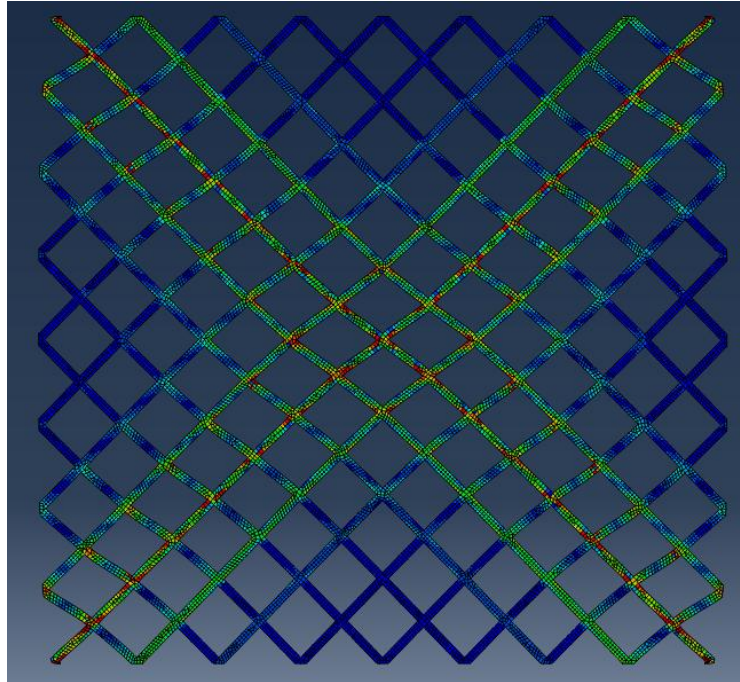


Figure 6: 8x8 Solid Lattice Cell Structure Under Displacement

In the force displacement graph for the 8x8 model we see major changes over the initial 4x4 graph. For this example, the elastic deformation region is between 0-1 meters, a 0.4-meter increase from the 4x4 lattice. The force at the elasticity limit is also almost 33% higher than the 4x4 matrix with a value of close to 4×10^7 newtons. Like the last lattice we do not have a lot of data past the yield point of the material as the displacement was not large enough to cause necking or fracture of the material during our trial.

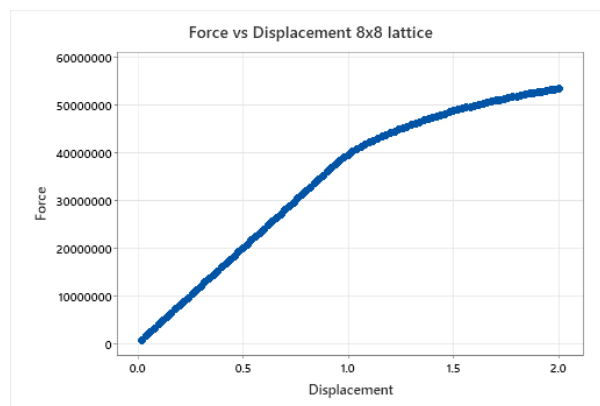


Figure 7: 8x8 Solid Lattice Force vs Displacement Graph

16x16 Solid Lattice

Like the findings in smaller lattice models, the 16x16 matrix shows a clear X-shaped pattern of stress distribution. This pattern becomes particularly evident because of the matrix's larger size, allowing for a broader perspective on stress transmission throughout the structure. The broadened scope shows a distinct accumulation of stress along the main diagonal axes, clearly demonstrating the pathways of force. The diamond-shaped unit cells in the matrix aid in this stress distribution by directing forces toward the central diagonal areas, where the greatest stress intensities occur. These areas, shouldering the majority of the burden, signify possible weak spots that might collapse under heightened displacement. The points of intersection along these diagonal axes persistently exhibit notable stress buildup, reflecting results observed in smaller lattice models and identifying them as crucial zones for structural stability.

The heightened dimensionality of the 16x16 matrix, while keeping a comparable displacement to smaller models, leads to a diminished overall deformation impact. The extra structural components present in the larger matrix enable a more even spread of applied forces, thus reducing localized stress concentrations in contrast to smaller lattices. This indicates that larger lattice formations of this design provide better load distribution and greater resistance to failure under comparable displacement circumstances. Nonetheless, the high-stress diagonal areas continue to be a major concern because of their ongoing vulnerability to strain-related deformation and possible failure. Consequently, although the larger matrix shows enhanced stress management, these essential diagonal areas require thorough attention in design and implementation.

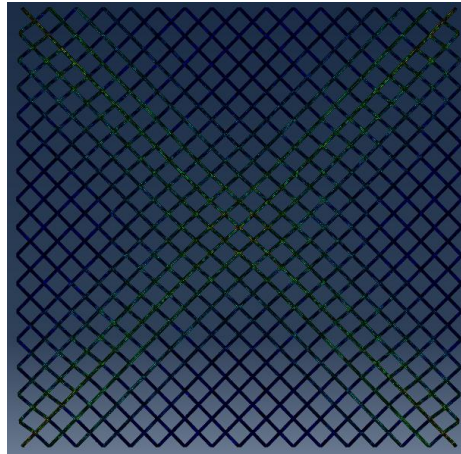


Figure 8: 16x16 Solid Lattice Cell Structure Under Displacement

In the force displacement graph for the 16x16 lattice, there is a notable rise in both the elastic deformation area and the force at the elasticity threshold when compared to smaller lattices. In this instance, the elastic deformation area reaches from 0 to about 2.0 meters, a significant rise from earlier models. The force at the elasticity threshold is significantly greater, nearing approximately 7×10^7 newtons. This suggests a significantly stronger and more durable material response in the 16x16 lattice. Like the earlier lattice examples, we do not see a distinct yield point or post-yield behavior like necking or breaking. This indicates that the applied displacement was inadequate to cause considerable plastic deformation in the material throughout the test.

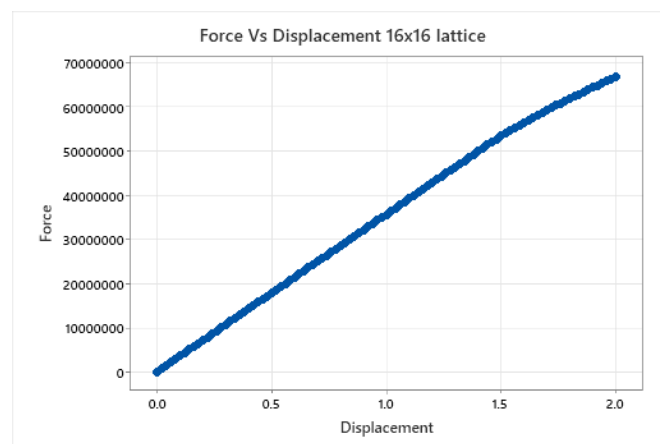


Figure 9: 16x16 Solid Lattice Force vs Displacement Graph

Solid Lattice Findings

As I advanced through the simulations, enlarging the lattice size showed considerable enhancements in stress dissipation and load distribution. Larger lattices, especially the 16x16, showed a more uniform distribution of applied forces, significantly lessening localized stress concentrations in contrast to the smaller 4x4 lattice. This suggested that larger lattice frameworks have improved structural stability and increased resistance to failure under comparable displacement scenarios.

The force-displacement charts offered additional useful information. I observed a distinct pattern: the area of elastic deformation expanded with lattice dimensions. The 16x16 lattice, for example, demonstrated a considerably greater elastic deformation area, suggesting a more robust and resilient material reaction. Regarding specific findings, the 4x4 lattice exhibited significant stress concentration mainly in the primary diagonal areas, with the elastic limit attained at about 0.6 meters of displacement. The 8x8 grid showed a more uniform stress distribution and a larger area of elastic deformation when compared to the 4x4. Ultimately, the 16x16 lattice demonstrated a notably expanded elastic deformation area and a considerably elevated force at the elastic limit, indicating a markedly stronger material performance.

Nevertheless, it was also clear that the displacement used in my simulations was inadequate to induce significant plastic deformation, like necking or fracture, in any of the lattice structures. This indicated that the information I gathered mainly concentrated on the elastic response and the early phases of plastic deformation, halting at the yield point. Further investigations with greater displacement will be essential to thoroughly examine the post-yield behavior and identify the ultimate stress and fracture points of these lattice frameworks.

In summary, my research emphasized the advantages of larger lattice sizes concerning better load distribution and increased structural integrity. The uniform "X" pattern of stress distribution and the pivotal importance of the diagonal element junctions were significant discoveries. Additionally, the expanding elastic deformation area with lattice dimensions signified a notable improvement in material strength and resilience. Although additional testing is required to completely grasp the post-yield behavior, my simulations established a strong basis for comprehending the structural behavior of solid lattices under displacement.

Wire lattice

In the 2D wire model simulations, employing beam elements with a radius of 0.1, I noticed a clear pattern of stress distribution when displacement was applied. Rather than a solid "X" formation, the stress focused along the beams' length, especially at the joints and areas of curvature within the structure. The structure, made up of interlinked wire-like beams, transmitted forces mainly through these beam components. The areas displaying warmer colors—yellow, orange, and red—represent regions of increased bending moment and axial stress, illustrating the beam's reaction to the load.

I also observed that the straight parts of the beams, distant from joints and curves, showed considerably lower stress levels, marked by cooler blue tones. This indicates that the forces exerted are not evenly spread but are focused on the beam's crucial structural locations, like joints and sections where the beam alters direction. The model's reaction to the load indicated that, although deformation took place, the structure stayed stable within the defined displacement boundaries, as shown by the lack of catastrophic failure.

I noticed an improvement in load distribution with a greater quantity of beams and connections in larger model sizes. The extra beam elements helped in distributing forces more efficiently, reducing the severity of localized stress concentrations at specific joints. Nevertheless, the joints and curved areas of the beams persisted in experiencing the greatest stress concentrations, rendering them crucial points for possible failure during heightened loading situations. This shows that even in a basic wireframe model, the geometric arrangement and connections of the beam elements are essential for defining the structural response.

4x4 Wire Lattice

In this 4x4 wire beam lattice configuration, I noticed heightened stress concentration mainly at the intersections and corners of the diamond-shaped cells, which experienced the greatest forces during deformation. The color gradient distinctly indicated that the connections of these beam elements faced the highest forces, especially in the red areas, implying possible critical points for structural failure. The linear sections of the beams, especially at their centers, encountered comparably low stress levels, as shown by the blue and green colors. These regions seemed to stay mostly untouched during peak load conditions, suggesting that they did not markedly impact the structural response in this loading situation. Importantly, even with the applied displacement, the structure deformed without showing any clear indications of total failure. Nonetheless, the regions under the greatest pressure, especially the vivid red areas, indicated locations where the substance was subjected to significant bending forces and longitudinal stresses. These localized stress points indicated that if the load increased further, these junctions would probably be the first to experience failure, possibly due to material yielding or breakage. In general, although deformation occurred, the structure preserved its integrity under the specified conditions; nonetheless, close attention must be given to the highly stressed joints to avert possible failure in prolonged or higher loading situations.

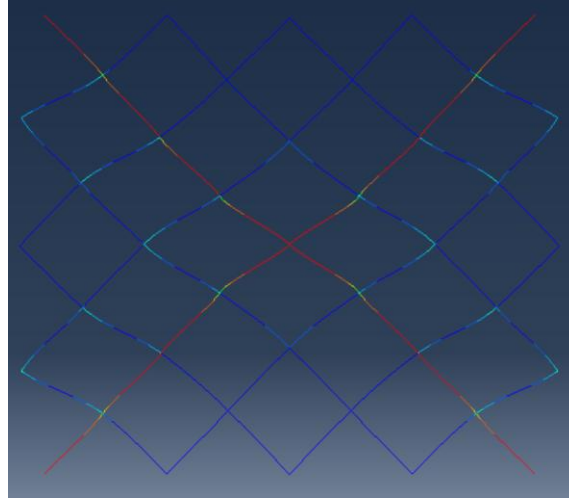


Figure 10: 4x4 Wire Lattice Cell Structure Under Displacement

Analyzing the graph I created for the 4x4 lattice, I observed an almost flawless linear relationship between force and displacement. When I raised the displacement, the force reacted with a steady, proportional rise, establishing that evident linear pattern. This indicates that the lattice operated completely within its elastic zone for the range of displacements I applied. There's no apparent bend or leveling of the line suggesting the beginning of yielding or plastic deformation. The structure appeared to be quite rigid and effectively withstood the applied load without displaying any notable non-linear behavior. The data points are tightly grouped along the line, reinforcing the reliability of the lattice's response and the precision of the measurements.

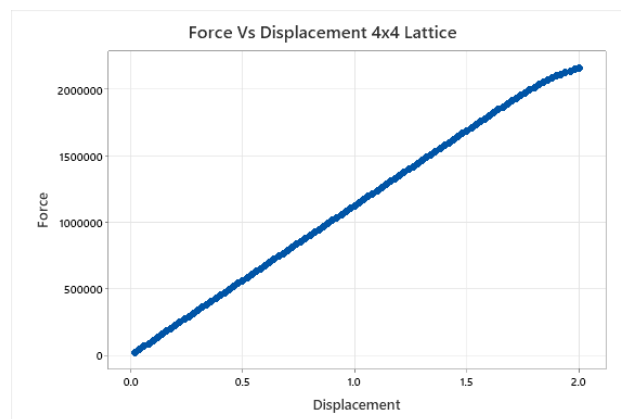


Figure 11: 4x4 Wire Lattice Force vs Displacement Graph

8x8 Wire Lattice

In this 4x4 wire beam section lattice setup, I again observed increased stress concentration primarily at the middle junction, which endured the most significant force during distortion. The color gradient clearly showed that this central connection experienced the greatest stress, particularly in the red zone, suggesting a potential critical point for structural collapse. The linear parts of the diagonal beams, particularly those further from the center, experienced relatively low stress levels, indicated by the blue and green hues. These areas appeared to remain largely unaffected during maximum load scenarios, indicating that they did not significantly influence the structural reaction in this loading context. Crucially, despite the applied load, the structure warped without displaying any obvious signs of complete failure. However, the area experiencing the most strain, particularly the bright red section, signified a place where the material faced considerable bending forces and long-axis stresses. This localized stress area suggested that if the load grew more, this central junction would likely be the first to fail, potentially due to material yielding or fracture. Overall, while deformation took place, the structure maintained its integrity under the given conditions; however, careful monitoring of the heavily stressed central joint is essential to prevent potential failure in extended or increased load scenarios.

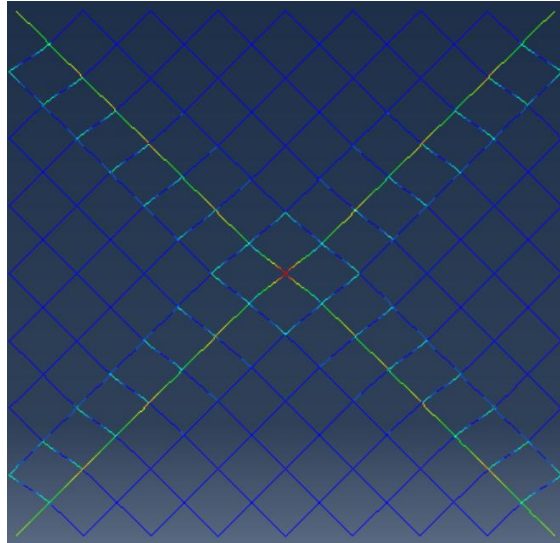


Figure 12: 8x8 Wire Lattice Cell Structure Under Displacement

In the graph of force versus displacement for the 8x8 lattice, I noticed a remarkably linear correlation. As displacement grew, the force increased proportionally, creating an almost flawless straight line. This indicates that the lattice stayed completely within its elastic zone throughout the examined displacement range. The steady, linear reaction suggests a rigid framework with little non-linear distortion. The closely packed data points enhance the precision and dependability of the measurements.

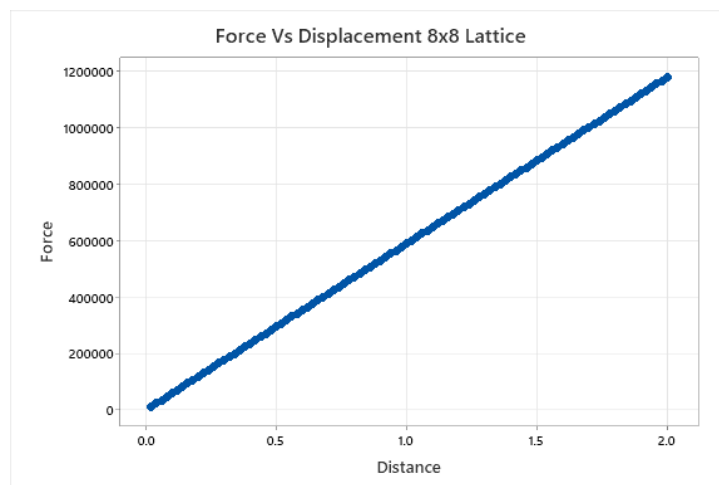


Figure 13: 8x8 Wire Lattice Force vs Displacement Graph

16x16 Wire Lattice

In this 16x16 wire beam lattice configuration, I observed a significant accumulation of stress at the central connection, bearing most of the applied load. The distinctive red tone at this point on the color gradient highlights its role as a potential failure point. Importantly, while the lattice's size increased, the stress remained concentrated at this central point, highlighting the design's efficiency in load transfer.

The diagonal beam sections, particularly those located farthest from the center, demonstrated minimal stress, indicated by blue and green hues. This suggests that these areas primarily act as stabilizers, whereas the diagonals function as the main load-bearing paths. Despite the exerted load, the structure maintained its integrity, displaying no indications of imminent failure. However, the intense red area at the center indicates increased bending and longitudinal stresses, implying it is likely the failure point under heavier loads.

The consistent stress concentration at the central node across various lattice sizes indicates an important design principle: the central node serves as a crucial load transfer restriction. Considering the lattice's measurements, a malfunction in this region would likely trigger a chain reaction of distortion. Therefore, it is crucial to closely monitor this important joint to prevent failure when exposed to

extended or increased loads. The 16x16 model highlights the importance of this central node as a key component for ensuring structural stability.

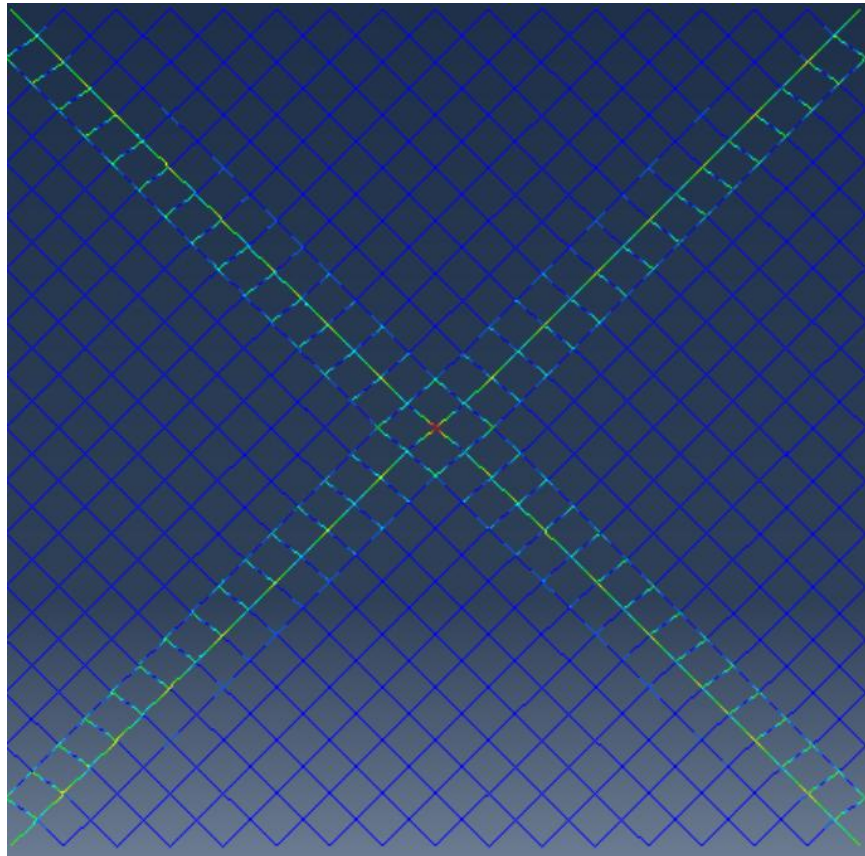


Figure 14: 16x16 Wire Lattice Cell Structure Under Displacement

In the force versus displacement graph for the 16x16 lattice, I noticed an impressively linear correlation. As I raised the displacement, the force corresponded with a steady, direct rise, forming an almost perfect straight line. This strongly indicates that the lattice stayed completely within its elastic deformation limits. The steady, straight-line response suggests a rigid framework with little non-linear distortion or yielding. The closely grouped data points along the line enhance the dependability and uniformity of the measurements.

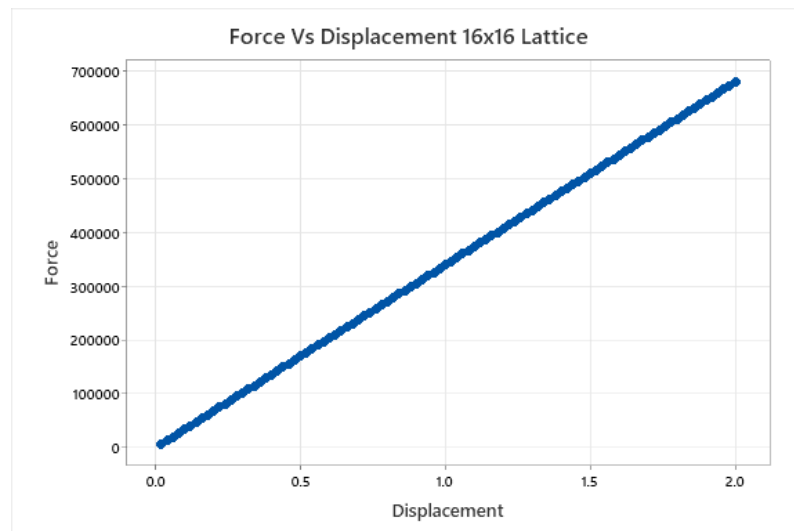


Figure 15: 16x16 Wire Lattice Force vs Displacement Graph

Overall Findings

This research examined the mechanical properties of 2-D diamond lattice frameworks subjected to applied displacement, exploring changes in lattice dimensions (4x4, 8x8, and 16x16). Main discoveries comprise:

- **Stress Distribution:** A uniform "X" pattern of stress distribution was noted across every lattice size, with peak concentrations found at the central intersection. This pattern emphasizes the effective load distribution system of the lattice, directing forces to the central node.
- **Size Effect:** Bigger lattices showed improved stress relaxation and load sharing. The 16x16 lattice, specifically, showed a more uniform stress distribution, decreasing localized stress concentrations when compared to smaller versions. This implies that a larger lattice size enhances structural integrity.
- **Force-Displacement Characteristics:** The force-displacement plots indicated a notable rise in the elastic deformation area as lattice size increased. The 16x16 lattice exhibited the greatest elastic deformation, suggesting enhanced material strength and rigidity.

Limitations and Future Experiments

This research provides important insights into the properties of 2-D diamond lattice structures, but it is crucial to acknowledge certain limitations. Using linear elastic material properties in the simulations restricts the analysis to a specific behavior range and may not fully depict the material's true response. Moreover, the two-dimensional nature of the models might not effectively depict the behavior of three-dimensional structures, which could exhibit more complex stress distributions and failure modes. Furthermore, the optimal loading conditions used in the simulations may not completely reflect the differences and imperfections found in real-world scenarios.

To enhance our understanding of lattice structures, I believe future research could examine these areas: Incorporate non-linear material properties:

- Incorporating non-linear material features, we can assess the effects of plasticity and material yielding on the lattice's behavior under more realistic loading conditions.
- Investigate 3-D lattice formations: Expanding the research to cover 3-D structures would provide enhanced understanding of their behavior and allow for a more accurate representation of practical applications.
- Analyze the impact of imperfections: It is crucial to explore how geometric imperfections and manufacturing tolerances influence the performance of the lattice to understand the dependability of these structures.
- Dynamic loading conditions: Investigating the response of the lattice to dynamic forces such as impacts or vibrations would provide valuable insights into its performance in real-world scenarios.

- Explore various lattice arrangements: Expanding the study to encompass different lattice configurations such as Kagome or octet truss lattices would enable a comparative assessment of their mechanical properties and identify the most suitable lattice designs for various uses.

By tackling these limitations and investigating these directions for future research, we can achieve a more profound comprehension of the mechanical properties of lattice structures and enhance their design and usage across different engineering disciplines.